AANet: An Ambiguity-Aware Network for Remote-Sensing Image Change Detection

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***Abstract*— Remote-sensing image change detection (CD) task plays an important role in land-use surveys, city construction investigations, and other vital industries. Recently, deep learning has become a mainstream method for this task due to its satisfactory performance in most cases. However, it often suffers from difficulties in dealing with ambiguous regions, where pseudo-changes happen or real changes are corrupted. In this article, we propose an ambiguity-aware network (AANet) to address the aforementioned issue. Specifically, our network first adopts convolutional layers to learn features from dual-temporal images. After that, an ambiguity refinement module (ARM) is designed to extract the ambiguity regions and then difference features are generated based on it. Considering that the scales of different changed objects vary, a weight rearrangement module (WRM) is proposed to fuse the difference features from different layers. To test the performance of our proposed model, we conduct experiments on three benchmark datasets, including SYSU-CD, SVCD, and LEVIR-CD. The experimental results show that our model can outperform several state-of-the-art models on all three datasets, which validates its effectiveness. The source code of our proposed model will be released at *https://github.com/KevinDaldry/AANet*.**

***Index Terms*— Ambiguity-aware network (AANet), change detection (CD), convolutional features, multiscale objects.**

I. Introduction

CHANGE detection (CD) is a task to detect changed objects of interest in the same area between dual or multiple periods. It has been widely used in multiple fields such as disaster assessment [1], land-use survey [2], and city construction investigation [3], helping us better understand the environment, detect potential anomalies, and facilitate decision-making.

To effectively detect the changed objects, much research has been conducted over the past decades. Generally, they can be divided into traditional methods and deep-learning-based methods. Most of the traditional methods [4], [5], [6] capture image changes based on mathematical and statistical approaches, which can be grouped into three main categories: the transformation-based approach, the algebra-based approach, and the classification-based approach. The first category of approach tends to apply change vector analysis [7], principal component analysis [8], multivariate alteration detection, and other techniques to enhance the changed features. The second category of approach exploits arithmetic strategies, such as ratio calculation [9], regression analysis, and difference normalization [10] to detect changes. The last category of approach attempts to use conventional machine-learning models like support vector machines [11] and K-means clustering for classifying changed and unchanged regions in a pixel-wise manner. However, such methods heavily rely on the quality of empirically designed features and have insufficient robustness under various situations.

Unlike the traditional methods, deep-learning-based methods can integrate feature extraction and CD within one single network to directly produce detection results in an end-to-end manner. Among all the deep-learning models, the convolutional neural network (CNN) becomes the mainstream structure because of its strong capability in image processing. One classical method uses CNNs as encoders to extract features from input images and the last-layer feature in the encoders is employed for CD [12], [13], [14], [15], [16]. However, such kind of methods make it difficult to detect changed objects with different scales. To solve this problem, Daudt et al. [17] exploited features from different layers in the encoder and used concatenation and subtraction operators to combine these multiscale features. Following that, many works [18], [19], [20], [21], [22], [23], [24] attempted to design different fusion strategies to improve the detection performance. Considering that CNNs focus on extracting local features, which cannot ideally capture global information, attention mechanism [25] was incorporated into CNN backbones in [26] to alleviate this issue. Feng et al. [27], showed that combining CNNs and global information from attention blocks can perform better in certain changed objects. Since then, many attempts [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38] have been conducted to adjust attention blocks within CNNs. In addition to constructing different network structures, there also exist some works aiming to design better loss functions for the CD task. For example, Chen et al. [39] designed a contrastive loss function that pulls the unchanged pixels closer while pushing the changed ones farther to constrain the learning process of their

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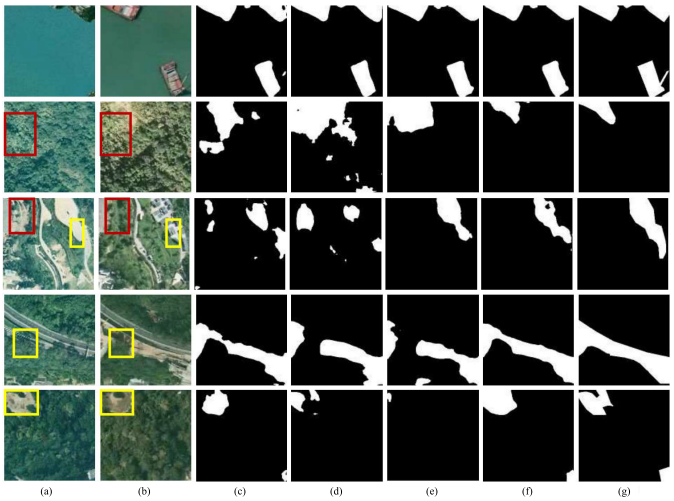

Fig. 1. Examples of some typical ambiguous situations. Red and yellow boxes represent pseudo-changes regions and shadow or shelter corrupted regions, respectively. Few existing models can deal with these cases simultaneously. (a) T1. (b) T2. (c) BIT. (d) DASNet. (e) DMINet. (f) Ours. (g) GT.

proposed network. Feng et al. [40] applied cross-entropy (CE) losses in the middle stages of the upsampling process to strictly supervise all the fusion processes, to relieve the misjudgment in generating CD masks.

To sum up, CNN-based CD methods can achieve satisfactory detection performance, especially on saliently changed or unchanged objects. Take the first row in Fig. 1 as an example, the saliently changed boats in the river can be detected by almost all state-of-the-art models. However, when there exist ambiguous regions, where pseudo-changes happen or real changes are corrupted in dual-temporal images, most of the existing deep-learning-based CD models suffer from difficulties. For example, as shown in the red box in Fig. 1, there are some pseudo-changes in the forest or grassland due to the season changes between the two images, but few of the existing models can deal with such issue. In addition, shadows from illumination changes and shelter caused by landscapes, shown in the yellow box in Fig. 1, will partly break the consistency of the changed objects, making the existing models cannot detect them entirely.

To address the aforementioned issues, we propose an ambiguity-aware network (AANet), which can pay more attention to processing ambiguity regions. Specifically, we take ResNet-18 [41] as the backbone to extract features from the input dual-temporal images. Then, for each pair of extracted convolution features, we use an ambiguity refinement module (ARM) to generate a difference feature. After that, we send all generated difference features into a weight rearrangement module (WRM) to adaptively combine them based on the observation that different changed objects have different scales. As shown in Fig. 1, our proposed network can achieve better detection results than the existing ones. In summary, our primary contributions can be concluded as follows.

1) Different from existing CD models, we propose an AANet, which can focus on the exploration of ambiguous regions. To achieve this goal, we design a plug-and-play ARM and use it to help extract different features from dual-temporal images.
2) Considering that the changed objects of interest often have different scales, we design a WRM to fuse different features from multiple convolution layers. Such a module is capable of adaptively learning the fusion weights of different layers.
3) We conduct comprehensive experiments on three widely used CD datasets, including SYSU-CD [42], SVCD [43], and LEVIR-CD [26]. The quantitative and qualitative results show that our proposed model can achieve better results than several state-of-the-art models.

The rest parts of our article are organized as follows. In Section II, we present our proposed model in detail. Experimental data and results are described in Section III. Finally, we give a conclusion of this article in Section IV.

II. Methodology

A. Overall Architecture

The flowchart of our proposed AANet is shown in Fig. 2. Assume $\{\mathbf{T}^1, \mathbf{T}^2\} \in \mathbb{R}^{3\times H\times W}$ represents a pair of dual-temporal images over the same area, where H and W denote the height and width of the two images, respectively. First, $\mathbf{T}^1$ and $\mathbf{T}^2$ are separately fed into ResNet-18, which consists of four residual blocks, to learn convolutional features $\mathbf{F}_i^1 \in \mathbb{R}^{(2^{(i-1)}\times C)\times(H/2^{i+1})\times(W/2^{i+1})}$ and $\mathbf{F}_i^2 \in \mathbb{R}^{(2^{(i-1)}\times C)\times(H/2^{i+1})\times(W/2^{i+1})}$, where $i \in \{1, 2, 3, 4\}$ is the index of the residual blocks, and C represents the channel numbers of the extracted features from the first residual block. After that, each feature pair $\{\mathbf{F}_i^1, \mathbf{F}_i^2\}$ is sent into ARM to extract the ambiguity region and then generate an enhanced difference feature $\mathbf{E}_i \in \mathbb{R}^{2C\times(H/2^{i+1})\times(W/2^{i+1})}$ based on it. Finally, the generated $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$, and $\mathbf{E}_4$ are adaptively combined together with the help of WRM to derive the CD mask.

B. Ambiguity Refinement Module

In the CD task, saliently changed or unchanged objects are easy to handle due to their obviousness. However, disrupted by multiple disturbances, many wanted changes are blurred, and irrelevant changes appear, causing severe ambiguity. Most existing models are capable of detecting the salient changes well but fail to give correct results in ambiguous regions. Therefore, it is critical to extract and deal with such regions. To this end, we design ARM following each pair of convolutional features $\{\mathbf{F}_i^1, \mathbf{F}_i^2\}$. As shown in Fig. 3, ARM is mainly comprised of two modules: the ambiguity and difference extractor (ADE) and ambiguity-guided difference refiner (AGDR). The purpose of ADE is to extract ambiguous regions and difference features. Based on the ambiguous regions, AGDR aims to refine the difference features.

Fig. 4 shows the detailed structure of ADE and AGDR. Specifically, to extract the ambiguity regions, 1×1 convolutional operators are first employed to reduce the feature dimensionality of $\mathbf{F}_i^1$ and $\mathbf{F}_i^2$, respectively. Then, a similarity matrix between $\mathbf{F}_i^1$ and $\mathbf{F}_i^2$ is derived by computing their cosine similarity. In this matrix, its values would be very small when

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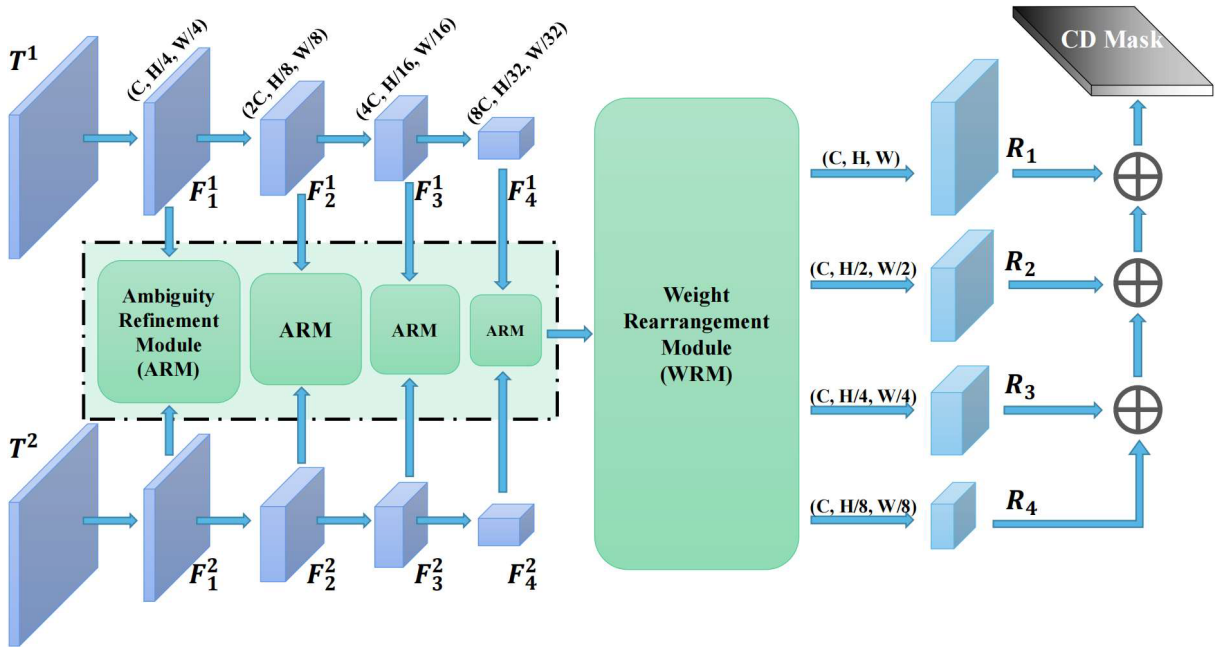

Fig. 2. Flowchart of our proposed AANet. The convolutional features extracted from the input dual-temporal images are separately fed into ARM to learn difference features. Then, these difference features are fused together with the help of WRM to generate the final CD mask.

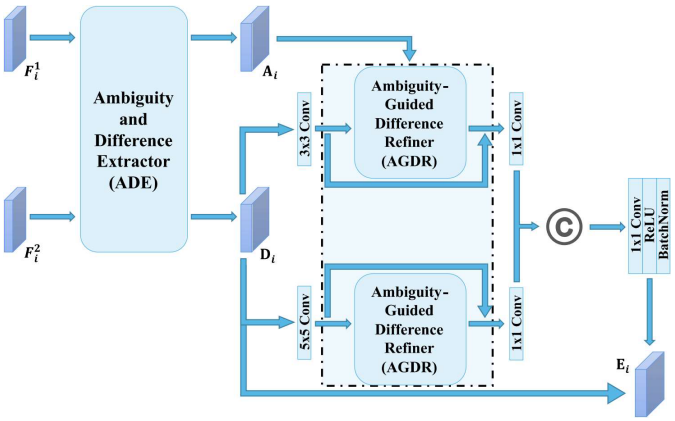

Fig. 3. Illustration of the proposed ARM. It first uses ADE to extract ambiguity and difference features. Then, the difference features are enhanced based on the help of ambiguity features through AGDR. "©" represents channel-wise concatenation.

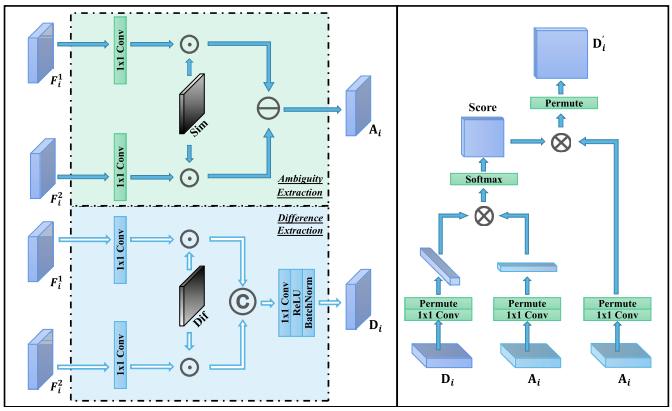

Fig. 4. Details of (Left) ADE and (Right) AGDR. "⊙", "⊗", and "⊖" represent the Hadamard product, standard matrix multiplication, and matrix subtraction, respectively.

the corresponding regions are saliently changed. So, if such matrix is multiplied with the dimensionality reduced $\mathbf{F}_i^1$ and $\mathbf{F}_i^2$, the saliently changed regions would be suppressed. After that, a simple subtraction operation is applied to suppress the unchanged regions. In summary, the aforementioned ambiguity extraction process can be represented as

$$\begin{aligned} \mathbf{Sim} &= \mathrm{Cosine}\left(\mathbf{F}_i^1, \mathbf{F}_i^2\right) \\ \mathbf{A}_i &= \left|\mathbf{Sim} \cdot \mathrm{Conv}\left(\mathbf{F}_i^1\right) - \mathbf{Sim} \cdot \mathrm{Conv}\left(\mathbf{F}_i^2\right)\right| \end{aligned} \tag{1}$$

where $\mathbf{Sim} \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})}$, *Cosine*, *Conv* and $\mathbf{A}_i \in \mathbb{R}^{2C\times(H/2^{i+1})\times(W/2^{i+1})}$ represent the similarity matrix, cosine function, 1×1 convolutional operator, and the ambiguity region, respectively.

In addition to $\mathbf{A}_i$, the difference feature $\mathbf{D}_i \in \mathbb{R}^{2C\times(H/2^{i+1})\times(W/2^{i+1})}$ is also needed because it is fundamental and necessary for the CD task. To acquire $\mathbf{D}_i$, a dissimilarity matrix $\mathbf{Dif} \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})}$ is directly derived according to **1-Sim** which indicates the salient differences, and then multiplied by $\mathbf{F}_i^1$ and $\mathbf{F}_i^2$ to get salient features $\{\mathbf{DF}_i^1, \mathbf{DF}_i^2\} \in \mathbb{R}^{(2^{(i-1)}\times C)\times(H/2^{i+1})\times(W/2^{i+1})}$. Finally, to combine $\mathbf{DF}_i^1$ and $\mathbf{DF}_i^2$ together, a concatenation operator along the channel dimension and a 1×1 convolutional operator are sequentially employed. The detailed process can be formulated as

$$\begin{aligned} \mathbf{Dif} &= \mathbf{1} - \mathbf{Sim} \\ \mathbf{DF}_i^1 &= \mathbf{Dif} \cdot \mathrm{Conv}\left(\mathbf{F}_i^1\right) + \mathbf{F}_i^1 \\ \mathbf{DF}_i^2 &= \mathbf{Dif} \cdot \mathrm{Conv}\left(\mathbf{F}_i^2\right) + \mathbf{F}_i^2 \\ \mathbf{D}_i &= \mathrm{Conv}\left(\mathrm{Cat}\left[\mathbf{DF}_i^1, \mathbf{DF}_i^2\right]\right) \end{aligned} \tag{2}$$

where *Cat* represents the concatenation operator.

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As discussed in Section I, there may exist changed objects in $\mathbf{A}_i$. Therefore, AGDR is designed to take advantage of such information to refine the difference features. In particular, $\mathbf{A}_i$ and $\mathbf{D}_i$ are separately put into 1×1 convolution to reduce their dimensions. Then, both $\mathbf{A}_i$ and $\mathbf{D}_i$ are spatially flattened to vectors and multiplied with each other. Via this operation, each pixel in $\mathbf{A}_i$ will produce a correlation score with pixels in $\mathbf{D}_i$ so that the global connection over all positions is stored and a weight map **Score** is generated. For the final refinement, **Score** is multiplied back to $\mathbf{A}_i$ to enhance the relevant regions and suppress the irrelevant ones, which can be formulated as

$$\begin{aligned} \mathbf{Score} &= \mathrm{Softmax}(\mathrm{Vec}(\mathrm{Conv}(\mathbf{D}_i))^T \times \mathrm{Vec}(\mathrm{Conv}(\mathbf{A}_i))) \\ \mathbf{D}'_i &= \mathbf{Score} \times \mathrm{Conv}(\mathbf{A}_i) \end{aligned} \tag{3}$$

where $\mathbf{D}'_i \in \mathbb{R}^{2C \times (H/2^{i+1}) \times (W/2^{i+1})}$ stands for the refined feature, *Vec* represents the vectorization operator, and the superscript "T" denotes the transpose operator.

Notably, to better capture the multiscale difference information, 3×3 and 5×5 convolutions are separately used on $\mathbf{D}_i$ before AGDR, and residual connections are used to stabilize the whole refinement. After AGDR, concatenation and 1×1 convolution operators are applied to fuse the generated multiscale features. The whole process can be formulated as

$$\begin{aligned} \mathbf{D}'_{3i} &= \mathrm{AGDR}(\mathrm{Conv}_{3\times3}(\mathbf{D}_i), \mathbf{A}_i) + \mathrm{Conv}_{3\times3}(\mathbf{D}_i) \\ \mathbf{D}'_{5i} &= \mathrm{AGDR}(\mathrm{Conv}_{5\times5}(\mathbf{D}_i), \mathbf{A}_i) + \mathrm{Conv}_{5\times5}(\mathbf{D}_i) \\ \mathbf{E}_i &= \mathrm{Conv}\left(\mathrm{Cat}\left[\mathrm{Conv}\left(\mathbf{D}'_{3i}\right), \mathrm{Conv}\left(\mathbf{D}'_{5i}\right)\right]\right) + \mathbf{D}_i \end{aligned} \tag{4}$$

where $\mathrm{Conv}_{n\times n}$ represents convolution operators whose kernel size is n, and $\mathbf{E}_i$ stands for the produced enhanced difference feature. At last, we send the multiple enhanced features $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \text{and } \mathbf{E}_4\}$ to WRM to adaptively fusing them together before producing the final CD mask.

C. Weight Rearrangement Module

Feature pyramid network (FPN) is a widely used structure to fuse features from different layers. It often upsamples the features in the latter layers to the same size as the former layers and combines them together to fuse the multiscale features. However, the traditional FPN considers that the importance of features from different stages is the same in the fusion phase [44], [45], leading to inaccurate CD results.

To solve this issue, as shown in Fig. 5, WRM is designed to redistribute the feature importance in a learnable manner before FPN. Specifically, global average pooling is first used on $\mathbf{E}_i$ to generate an averaged feature representation $\mathbf{V}_i \in \mathbb{R}^{2C}$. These representations are then concatenated to form a weight vector, which contains the channel-wise information from $\mathbf{E}_i$. Through a linear projection, the weight vector is globally fused to generate the redistributed weight sequence $\mathbf{W} \in \mathbb{R}^4$. Such a process can be represented as

$$\begin{aligned} \mathbf{V}_i &= \mathrm{GAP}(\mathbf{E}_i) \\ \mathbf{W} &= \mathrm{Linear}(\mathrm{Cat}[\mathbf{V}_i, i \in \{1, 2, 3, 4\}]) \end{aligned} \tag{5}$$

where *GAP* stands for the global average pooling operator and *Linear* means linear projection.

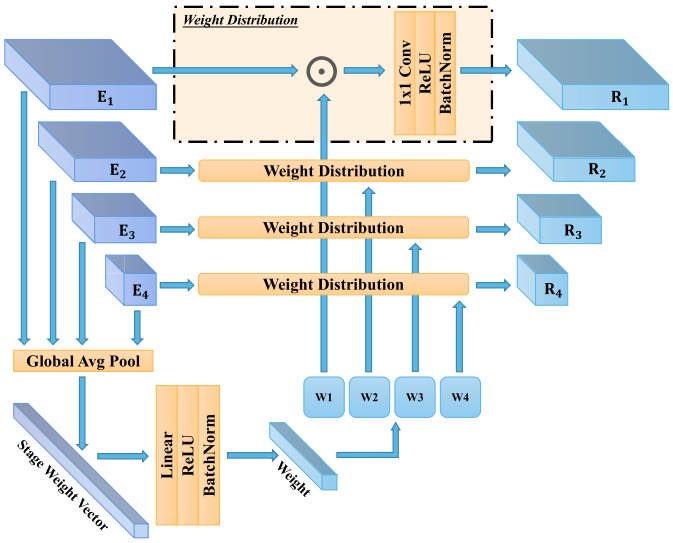

Fig. 5. Details of the WRM module.

After acquiring corresponding weight w_i from $\mathbf{W}$, the output $\mathbf{E}_i$ is re-weighted via dot product and 1×1 convolution is used to get the output feature $\mathbf{R}_i \in \mathbb{R}^{C \times (H/2^{i-1}) \times (W/2^{i-1})}$

$$\mathbf{R}_i = \mathrm{Conv}(w_i \cdot \mathrm{Upsample}_{\times 4}(\mathbf{E}_i)) \tag{6}$$

where $\mathrm{Upsample}_{\times 4}$ represents upsampling the spatial dimension of input features by four times via a bi-linear interpolation operator. Finally, all $\mathbf{R}_i$, $i \in \{1, 2, 3, 4\}$, are put into a standard FPN and produce the final CD mask.

D. Loss Function

In addition to network structures, loss function also plays an important role in deep-learning models. Considering that the ratio of changed objects in CD tasks is often less than unchanged ones, which makes the changed objects not fully learned, we adopt dice loss [46] to alleviate this issue, because it can pay more attention to the regions of interest. However, dice loss cannot effectively learn the unchanged objects. Thus, as compensation, we select CE loss along with dice loss to build a hybrid loss, which can be described as

$$\mathbf{L} = k \cdot \mathbf{L}_{\mathrm{CE}} + (1 - k) \cdot \mathbf{L}_{\mathrm{DICE}} \tag{7}$$

where $\mathbf{L}_{\mathrm{CE}}$ and $\mathbf{L}_{\mathrm{DICE}}$ represent the cross-entropy loss and the dice loss, respectively, k is a hyperparameter to balance the contributions between them. $\mathbf{L}_{\mathrm{CE}}$ and $\mathbf{L}_{\mathrm{DICE}}$ are calculated as

$$\begin{aligned} \mathbf{L}_{\mathrm{CE}}(\hat{\boldsymbol{y}}, \boldsymbol{y}) &= \boldsymbol{y} \cdot \log(\hat{\boldsymbol{y}}) + (\mathbf{1} - \boldsymbol{y}) \cdot \log(\mathbf{1} - \hat{\boldsymbol{y}}) \\ \mathbf{L}_{\mathrm{DICE}}(\hat{\boldsymbol{y}}, \boldsymbol{y}) &= \mathbf{1} - \frac{2 \cdot \boldsymbol{y} \cdot \hat{\boldsymbol{y}}}{\|\boldsymbol{y}\| + \|\hat{\boldsymbol{y}}\|} \end{aligned} \tag{8}$$

where $\|\cdot\|$ stands for L1-norm, $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}$ represent the ground-truth and the predicted results, respectively.

III. Experiments

A. Data Description

To test the performance of our proposed model, we conduct comprehensive experiments on three widely used CD datasets.

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1) *SYSU-CD:* It contains 20 000 pairs of aerial images over Hong Kong between 2007 and 2014. The spatial resolution and size of each image are 0.5 m, and 256 × 256, respectively. As a prosperous port, Hong Kong enjoyed rapid construction in ports, sea routes as well as oceanic and coastal projects during these seven years. Accordingly, this dataset focuses on change instances of high-rise towels, which are hard to mark because of the disturbance of deviation and shadow, and related changes around ports along with other major changes such as vegetation and road expansion. The same as the settings in [42], we also divide the whole dataset into 12 000 training pairs, 4000 validation pairs, and 4000 testing pairs.
2) *SVCD:* It is originally comprised of seven pairs of images shot by Google Earth during different seasons, aiming to focus on human land-use changes. The spatial resolutions range between 0.03 and 1 m. Following the work in [47], each image is cropped into several image patches with a size of 256 × 256. All of the image patches are divided into the training set, the validation set, and the testing set, whose sample numbers are 10 000, 2998, and 3000, respectively.
3) *LEVIR-CD:* It captures newly constructed and destructed building changes in Texas, the USA, between 2002 and 2018 by Google Earth. The total number of image pairs is 637, and the spatial size of each image is 1024 × 1024 pixels. According to the work in [26], each image is cropped into many patches whose size is 256 × 256. The number of training, validation, and testing samples is set to 7120, 1024, and 2048, respectively.

B. Implementation Details

All the models are implemented with PyTorch on a single NVIDIA Geforce RTX 3090 24G graphic card. To optimize our proposed network, we use stochastic gradient descent (SGD) with an initial learning rate setting of 0.1. During the training process, we let the learning rate decay with a power of 0.005 in every 100 iterations. The batch size is set to 32 to maximize the use of graphic sources and the number of epochs is 100. To fully evaluate the performance of each model, we use five metrics, including Precision (Pr), Recall (Re), F1 Score, Overall Accuracy (OA), and Intersection-of-Union (IoU). They are calculated as follows:

$$\begin{aligned} \text{Pr} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{Re} &= \frac{\text{TP}}{\text{TP}+\text{TN}} \\ F1\ \text{Score} &= \frac{2\times\text{TP}}{2\times\text{TP}+\text{FP}+\text{FN}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \end{aligned} \tag{9}$$

where TP and TN, respectively, represent the number of matched changed and unchanged pixels on CD masks, while FP and FN, respectively, denote the number of false alarms of changed and unchanged pixels.

TABLE I
DETECTION PERFORMANCE ON THE SYSU-CD DATASET

Method	Pr	Re	F1	OA	IoU
FC-EF	82.26	65.48	72.92	88.53	57.38
FC-Siam-conc	84.05	68.78	75.65	89.56	60.84
FC-Siam-diff	**90.25**	52.28	66.20	87.41	49.48
STANet	88.33	62.21	73.00	89.15	57.48
BIT	81.81	71.66	76.40	89.56	61.82
DASNet	87.57	71.96	79.00	90.98	65.29
DMINet	87.53	74.83	80.68	91.55	67.62
Ours	87.52	**78.15**	**82.57**	**92.22**	**70.31**

C. Comparisons With State-of-the-Art Models

We compare our proposed model with several state-of-the-art models, including FC-EF [17], FC-Siam-Conc [17], FC-Siam-Diff [17], STANet [26], BIT [36], DASNet [39], and DMINet [40]. To make a fair comparison, we re-implement them according to their open-source codes.

The experimental results achieved by different models on the SYSU-CD dataset are listed in Table I, where bold fonts represent the best results. From this table, it can be observed that although FC-Siam-diff achieves the highest Pr value, its Re value is the lowest, leading to much worse detection performance than other models in terms of F1, OA, and IoU. In comparison with plain CNN models (i.e., FC-EF, FC-Siam-conc, and FC-Siam-diff), STANet and BIT obtain better F1, OA, and IoU values due to the incorporation of attention mechanism. However, their performance is still not high, because they easily yield false-alarm results on ambiguous or inconspicuous objects. DASNet and DMINet can alleviate such issues by designing suitable loss functions, thus improving the detection performance of STANet and BIT. Compared to all of the above models, our proposed model can achieve the best Re, F1, OA, and IoU values, because it can well deal with the ambiguity regions. In addition to quantitative comparisons, Fig. 6 also qualitatively demonstrates the detection results of different models in some instances. In this figure, the red and green colors represent FP and FN results, respectively. It is shown that the seven comparison models tend to produce results with incomplete regions. Instead, our proposed model can overcome this issue and produce semantically consistent regions in most cases, which further certifies its effectiveness.

Tables II and III compare the detection performance of different models on the SVCD dataset and the LEVIR-CD dataset, respectively. Most of the changed objects in these two datasets are relatively easier to detect than the ones in the first dataset, so most of the models achieve significantly better performance. Specifically, on the SVCD dataset, our proposed model gets better results than the other models in all indicators. Although STANet obtains the best Pr value than other models on the LEVIR-CD dataset, its Re value is the worst, making the overall indicators, especially F1,

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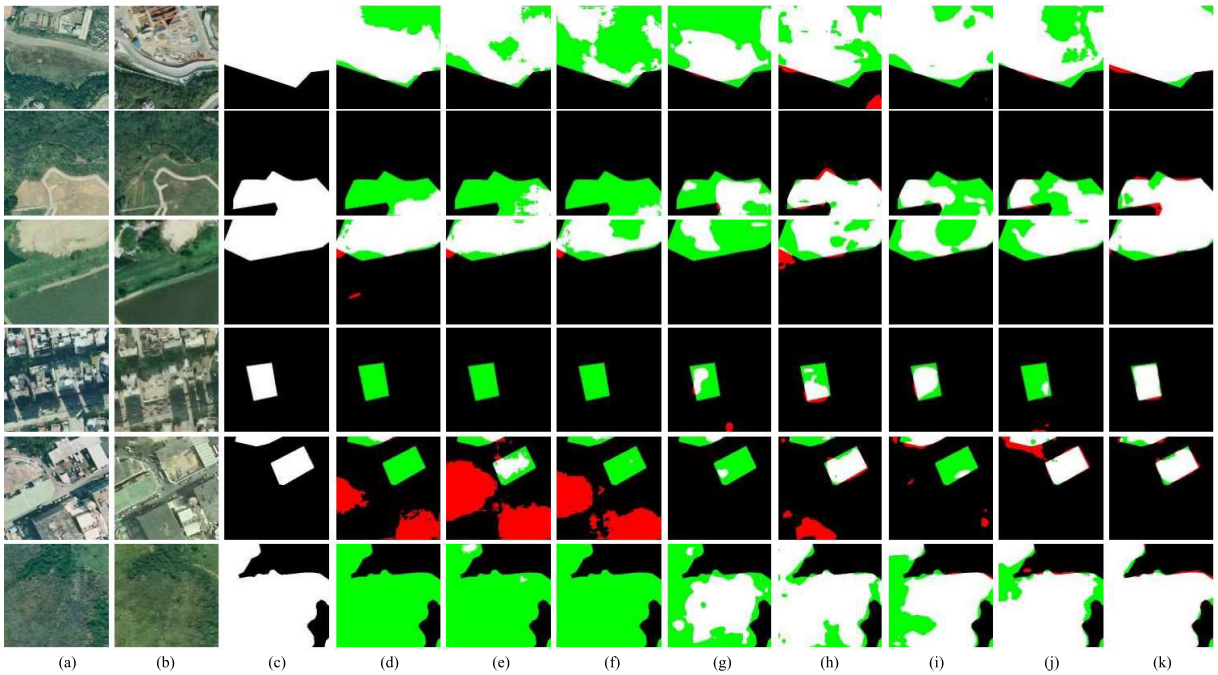

Fig. 6. Visualization results achieved by different models on some instances chosen from the SYSU-CD dataset. The red and green colors represent FP and FN results, respectively. (a) T1. (b) T2. (c) GT. (d) FCEF. (e) FCSC. (f) FCSD. (g) STANet. (h) BIT. (i) DASNet. (j) DMINet. (k) Ours.

TABLE II
DETECTION PERFORMANCE ON THE SVCD DATASET

Method	Pr	Re	F1	OA	IoU
FC-EF	85.15	52.36	64.84	93.30	47.98
FC-Siam-conc	81.72	60.47	69.51	93.74	53.26
FC-Siam-diff	90.07	54.65	68.02	93.94	51.45
STANet	95.93	91.12	93.47	98.50	87.73
BIT	96.06	91.97	93.97	98.61	88.63
DASNet	96.09	94.35	95.21	98.88	90.86
DMINet	96.21	93.85	95.02	98.84	90.51
Ours	**96.22**	**95.43**	**95.82**	**99.02**	**91.98**

TABLE III
DETECTION PERFORMANCE ON THE LEVIR-CD DATASET

Method	Pr	Re	F1	OA	IoU
FC-EF	80.64	86.80	83.61	98.27	71.83
FC-Siam-conc	86.48	83.41	84.92	98.49	73.78
FC-Siam-diff	81.46	**90.88**	85.91	98.64	75.30
STANet	**97.13**	72.68	83.14	98.50	71.75
BIT	92.17	86.43	89.21	98.93	80.52
DASNet	90.73	86.93	88.79	98.88	79.84
DMINet	91.87	88.45	90.13	99.01	82.03
Ours	92.68	89.11	**90.86**	**99.09**	**83.25**

and IoU, not high. Similarly, FC-Siam-diff achieves the highest Re but low Pr values. So, its F1, OA, and IoU are inferior to DASNet and DMINet. In comparison with them, our proposed model can achieve the second-best Pr and Re values, and the highest F1, OA, and IoU values. Figs. 7 and 8 show the detection results of some instances on the SVCD dataset and the LEVIR-CD dataset, respectively. For these instances, it can be observed that the results obtained by our proposed model are more consistent in comparison with their corresponding GT than the other models.

In addition, we also demonstrate the detection results of different models in the ambiguous situations extracted by ARM. As shown in Fig. 9, when dual-temporal landscapes tend to be very similar or complex, the three comparison models can hardly produce ideal detection results but our proposed model can. Specifically, when landscapes seem to be weak or nearly no changes, as shown in the first three rows in Fig. 9, BIT, DASNet, and DMINet are incapable of properly detecting these inconspicuous difference between dual-temporal features so that they miss or misjudge these targets. However, since the difference information is refined by extracted ambiguities, our proposed model can capture these targets and produce more accurate results. When landscapes are complex and have various changed objects, compared with BIT, DASNet, and DMINet, our proposed model can produce more complete masks because the changed information in our extracted ambiguity map is used to compensate for the learned difference features.

D. Ablation Study

As presented in Section II, our proposed model is mainly comprised of three modules (i.e., ADE, AGDR, and WRM). To test the effectiveness of them, we conduct an ablation study on all three datasets. The experimental results are demonstrated in Table IV, where "✓" denotes that the corresponding module is adopted. First of all, we remove all three modules and use the remainder of ResNet-18 as well as FPN to construct a *Baseline* model. Then, we gradually add each

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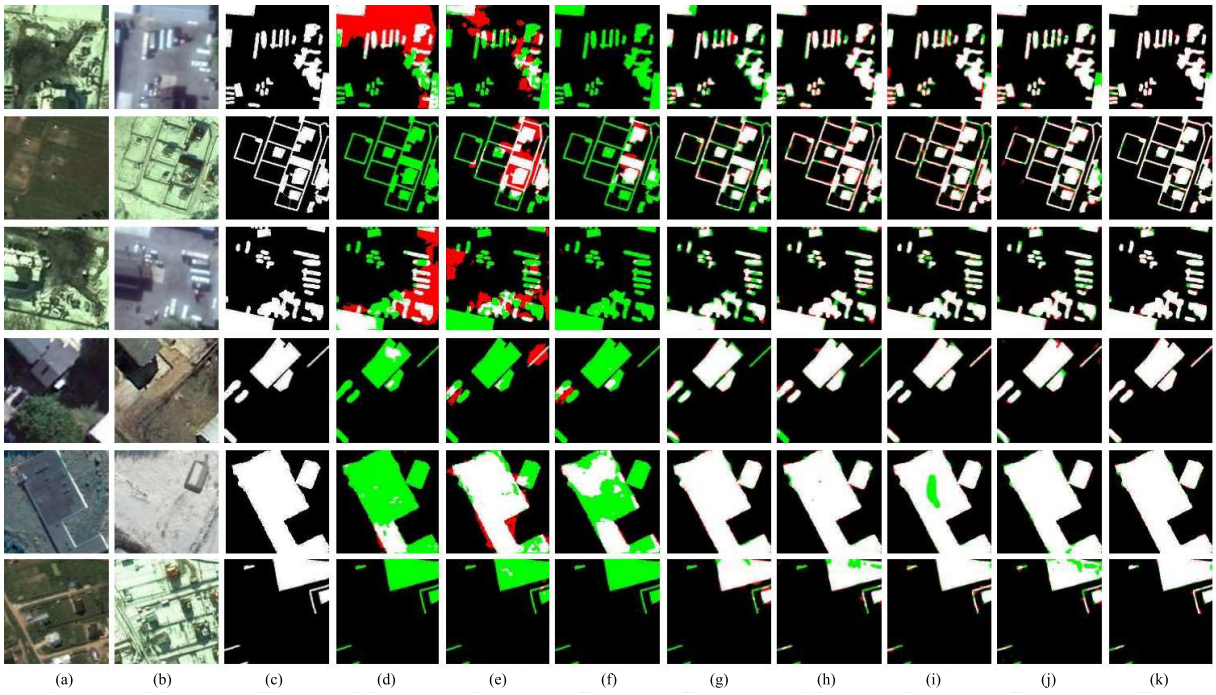

Fig. 7. Visualization results achieved by different models on some instances chosen from the SVCD dataset. The red and green colors represent FP and FN results, respectively. (a) T1. (b) T2. (c) GT. (d) FCEF. (e) FCSC. (f) FCSD. (g) STANet. (h) BIT. (i) DASNet. (j) DMINet. (k) Ours.

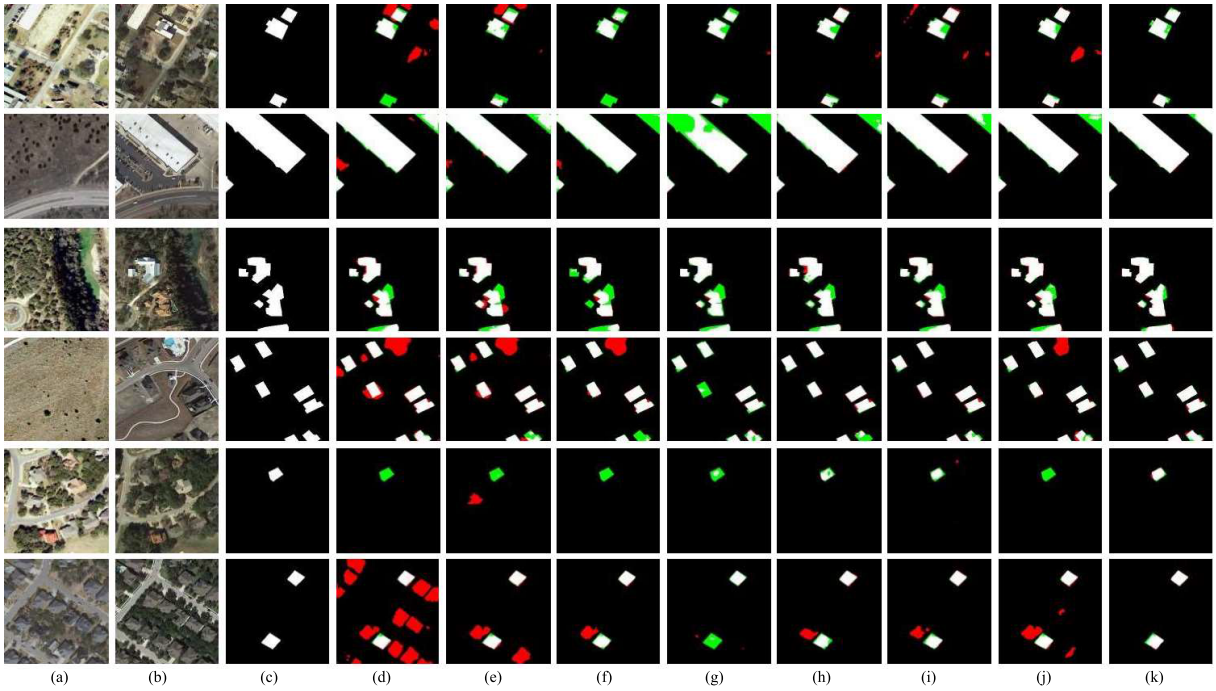

Fig. 8. Visualization results achieved by different models on some instances chosen from the LEVIR-CD dataset. The red and green colors represent FP and FN results, respectively. (a) T1. (b) T2. (c) GT. (d) FCEF. (e) FCSC. (f) FCSD. (g) STANet. (h) BIT. (i) DASNet. (j) DMINet. (k) Ours.

module into it. As can be seen in Table IV, when WRM is added into *Baseline*, both F1 and IoU are improved. When WRM and ADE are added into *Baseline*, we adopt an addition operator to combine the ambiguity and difference information extracted from ADE. Such a model makes a higher improvement than using WRM only in terms of F1 and IoU. After that, when the addition operator is replaced by AGDR, our proposed model can further boost the detection performance. Fig. 10 visualize the detection results with or without using ARM (i.e., both ADE and AGDR), where "w" and "w/o" represents ARM is and is not adopted in our proposed model, respectively. It is obvious that using ARM can produce more complete objects and filter out the pseudo-changed objects.

Additionally, to evaluate the effects of different fusion methods on combining $\mathbf{A}_i$ and $\mathbf{D}_i$, we compare AGDR with

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TABLE IV
ABLATION STUDY OF OUR PROPOSED MODEL ON THREE DATASETS

Modules				F1 Score			IoU		
Baseline	WRM	ADE	AGDR	SYSU	SVCD	LEVIR	SYSU	SVCD	LEVIR
✓				79.77	94.29	89.15	66.34	89.19	80.42
✓	✓			80.06	94.68	89.74	66.76	89.90	81.38
✓	✓	✓		81.61	95.56	90.41	68.93	91.49	82.51
✓	✓	✓	✓	**82.57**	**95.82**	**90.86**	**70.31**	**91.98**	**83.25**

TABLE V
EFFECTS OF DIFFERENT FUSION METHODS

Fusion Method	F1 Score			IoU		
	SYSU	SVCD	LEVIR	SYSU	SVCD	LEVIR
No Fusion	80.53	94.71	89.84	67.41	89.95	81.55
Addition Fusion	81.61	95.56	90.41	68.93	91.49	82.51
Convolutional Fusion	80.67	95.26	90.56	67.61	90.95	82.76
AGDR	**82.57**	**95.82**	**90.86**	**70.31**	**91.98**	**83.25**

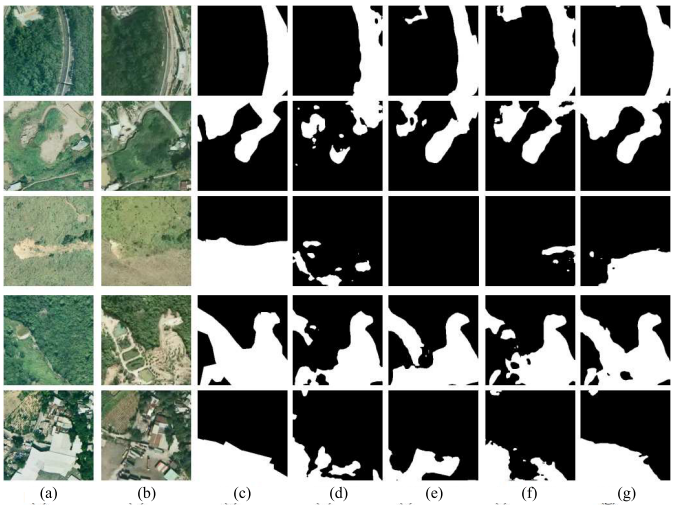

Fig. 9. Visualization results achieved by different models in ambiguous situations. (a) T1. (b) T2. (c) GT. (d) BIT. (e) DASNet. (f) DMINet. (g) Ours.

two widely used fusion methods, including addition fusion and convolutional fusion. For the addition fusion method, $\mathbf{A}_i$ and $\mathbf{D}_i$ are directly added together. For the convolutional fusion method, $\mathbf{A}_i$ and $\mathbf{D}_i$ are first concatenated together, and then 1×1 convolutional operator is adopted. Table V reports the detection performance of different fusion methods on three datasets, where the first row represents that only $\mathbf{D}_i$ is extracted and there is no fusion operators. From this table, it can be observed that all three fusion methods can improve the detection performance when compared to the individual use of $\mathbf{D}_i$, which indicates the importance of extracting ambiguous regions. More importantly, in comparison with the other two fusion methods, our proposed AGDR achieves superior performance on all datasets, which validates its effectiveness and robustness.

To make an intuitive understanding of $\mathbf{A}_i$ and $\mathbf{D}_i$, we visualize some intermediate results in our proposed model in Fig. 11.

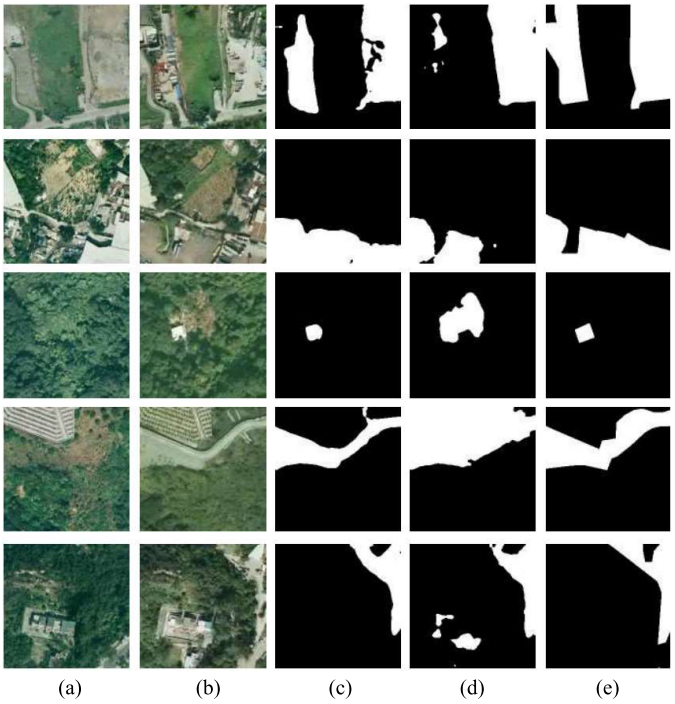

Fig. 10. Visualization results of our proposed model with (w) and without (w/o) using ARM. (a) T1. (b) T2. (c) w ARM. (d) w/o ARM. (e) GT.

Note that the hotter colors in $\mathbf{A}_i$ indicate more ambiguous regions, while those in the other maps represent changed regions with higher probability. From this figure, it can be observed that for saliently changed objects, as shown in Fig. 11(a), $\mathbf{D}_i$ is good enough to detect them, and there exist fewer ambiguity regions in $\mathbf{A}_i$. On the contrary, for the complex scenario in Fig. 11(b), $\mathbf{D}_i$ cannot detect most of the changed regions accurately. However, $\mathbf{A}_i$ can pay attention to the ambiguous regions and help $\mathbf{D}_i$ find the changed objects, as shown in $\mathbf{E}_i$.

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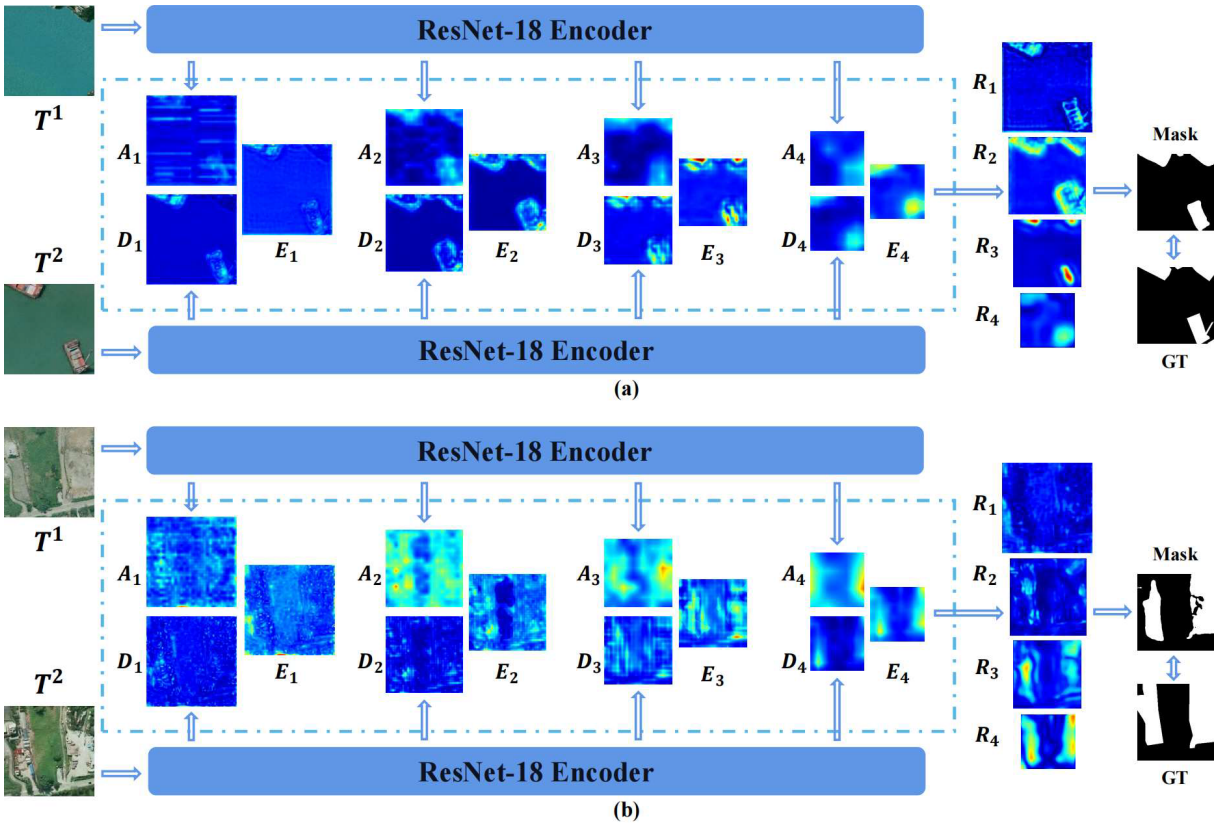

Fig. 11. Visualization of some intermediate results in our proposed model. (a) Intermediate results of a saliently changed object. (b) Intermediate results of a complex scenario where ambiguous regions exist.

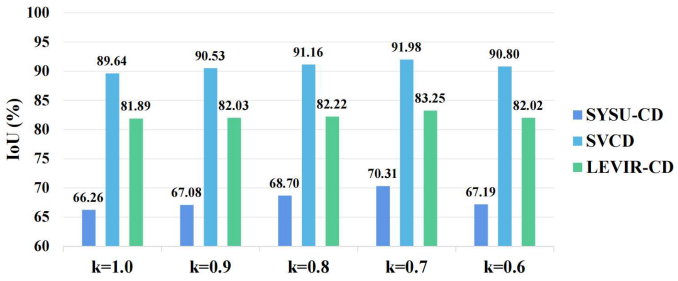

Fig. 12. Effects of different k values on the detection performance.

As presented in (7), there exists a hyperparameter k in our proposed loss function. To choose a relatively good value for it, we change its values from a candidate set $\{1.0, 0.9, 0.8, 0.7, 0.6\}$ and calculate the corresponding detection performance in terms of IoU. Fig. 12 shows the detailed results achieved by different k values. It can be observed that when k is decreased from 1 to 0.7, the IoU values on three datasets are gradually increased. However, when k is changed from 0.7 to 0.6, the IoU values are decreased. Therefore, it is reasonable to set the k value to 0.7 on all three datasets.

IV. Conclusion

In this article, an AANet was proposed for remote-sensing image CDs. Different from most existing deep-learning models, our proposed network focuses on exploring the ambiguity regions in dual-temporal images, which is a challenging but important issue. To this end, we designed an ambiguity and difference extractor to extract the ambiguity regions and then proposed an AGDR to learn difference features. In addition, considering that the changed objects often have different scales, a WRM was designed to combine the difference features learned from different layers. To evaluate the performance of our proposed model, three widely used datasets were employed to conduct experiments. Experimental results show that our proposed model can achieve better detection results than several state-of-the-art models. Also, the ablation study experiments were conducted and the effectiveness of each module in our proposed model is sufficiently validated.

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